PH.45320 **FALL 2026**

Static Cosmology

Lecture 13: Age Lower Bounds in Eternal Equilibrium

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Overview

Independent cooling, nuclear, and rotational clocks provide lower bounds on the duration of equilibrium.

1 Observational setup

Source coordinates, observing band, selection function, and calibration state are fixed at the outset.

$$p(\tau|D) \propto L_{cool} L_{nuc} L_{rot}\, p(\tau)$$

2 Transport or equilibrium equation

The governing equation is solved first in a homogeneous limit and then with the stated spatial dependence.

$$n(\tau) = RP_{surv}(\tau)$$

3 Connection to data

Each model quantity is tied to a measured flux, spectrum, angle, count, or temperature.

$$T_{background} > max(\tau_{relax})$$

4 Degeneracies

Calibration, population, and medium parameters are varied separately to expose their degeneracies.

5 Example

The worked example carries units and uncertainty from the input catalog to the reported result.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Jeong, ch. 13
- Stellar Age Lower-Bound Catalog
- Equilibrium Timescale Standard